

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_diversity_failure_replay.
- Best TSPLIB holdout gap: phase6_diversity_weighted_replay.
- Best synthetic holdout gap: phase6_failure_replay.

Run Metadata

- run_name: run_20260518_231209_c
- started_at_local: 2026-05-18 23:12:09
- finished_at_local: 2026-05-18 23:32:53
- duration_hhmm: 00:21
- duration_seconds: 1243.645
- seed_offset: 2000
- replicate_label: c
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.086291	0.010343	0.058674	0.236772	0.162469	0.0	0.764505	0.76	0.036986
phase6_random_replay	random	score_only	False	0.210832	0.012044	0.138545	0.236772	0.153461	0.19457	0.0	0.76	0.0
phase6_failure_replay	failure	score_only	False	0.143482	0.0	0.091307	0.171206	0.27492	0.222081	0.0	0.76	0.0

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.114249	0.0	0.072704	0.236772	0.16683	0.19457	0.805692	0.76	0.013487
phase6_stratified_random_replay	stratified_random	score_only	False	0.141226	0.0	0.089871	0.236772	0.186949	0.221162	0.853725	0.76	0.044302
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.084761	0.003941	0.055372	0.236772	0.16044	0.157119	0.817261	0.76	0.04933
phase6_residual_failure_replay	residual_failure	score_only	False	0.197162	0.0	0.125467	0.158073	0.331945	0.367262	0.0	0.76	0.0
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.086663	0.0	0.055149	0.171206	0.230022	0.174426	0.814708	0.76	0.059496
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.134684	0.003941	0.087141	0.171206	0.3043	0.17767	0.645429	0.76	0.049276

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.086291, synthetic 0.010343, combined 0.058674.
- Accepted-epoch count 3, mean accepted code novelty 0.764505, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.162469, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.036986 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.086291 across 7 instances; family means: ch=0.068918, kroD=0.037287, pcb=0.155796, pr=0.111484, rd=0.069785, st=0.091852.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3178, "expected_gap": 0.40349, "family": "a", "name": "a280", "optimality_gap": 0.232261, "residual_gap": -0.171229}, {"best_known_cost": 7542, "cost": 8291, "expected_gap": 0.240918, "family": "berlin", "name": "berlin52", "optimality_gap": 0.099311, "residual_gap": -0.171229}]

-0.141607}, {"best_known_cost": 14379, "cost": 15300, "expected_gap": 0.32496, "family": "lin", "name": "lin105", "optimality_gap": 0.064052, "residual_gap": -0.260908}]

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.210832, synthetic 0.012044, combined 0.138545.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.153461, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133388, mean selected residual gap -0.016771.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.210832 across 7 instances; family means: ch=0.195686, kroD=0.215835, pcb=0.267833, pr=0.315646, rd=0.176991, st=0.108148.
- Panel synthetic_holdout mean gap 0.012044 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.048176, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3347, "expected_gap": 0.40349, "family": "a", "name": "a280", "optimality_gap": 0.29779, "residual_gap": -0.1057}, {"best_known_cost": 14379, "cost": 18560, "expected_gap": 0.321761, "family": "lin", "name": "lin105", "optimality_gap": 0.290771, "residual_gap": -0.03099}, {"best_known_cost": 7542, "cost": 8325, "expected_gap": 0.236622, "family": "berlin", "name": "berlin52", "optimality_gap": 0.103819, "residual_gap": -0.132803}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.143482, synthetic 0.0, combined 0.091307.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.27492, size bias 0.120322, and failure concentration 0.222081.
- Replay selection diversity 0.176857, mean selected expected gap 0.314562, mean selected residual gap -0.010364.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.143482 across 7 instances; family means: ch=0.091634, kroD=0.177374, pcb=0.206901, pr=0.236171, rd=0.175474, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3358, "expected_gap": 0.403645, "family": "a", "name": "a280", "optimality_gap": 0.302055, "residual_gap": -0.10159}, {"best_known_cost": 7542, "cost": 9372, "expected_gap": 0.233148, "family": "berlin", "name": "berlin52", "optimality_gap": 0.242641, "residual_gap": 0.009493}, {"best_known_cost": 629, "cost": 756, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.201908, "residual_gap": -0.052782}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.114249, synthetic 0.0, combined 0.072704.

- Accepted-epoch count 3, mean accepted code novelty 0.805692, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.16683, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132188, mean selected residual gap 0.004372.
- Adaptation efficiency 0.013487 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 8, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.114249 across 7 instances; family means: ch=0.061092, kroD=0.237767, pcb=0.221041, pr=0.113721, rd=0.082807, st=0.022222.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19562, "expected_gap": 0.323194, "family": "lin", "name": "lin105", "optimality_gap": 0.360456, "residual_gap": 0.037262}, {"best_known_cost": 629, "cost": 788, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.252782, "residual_gap": -0.000636}, {"best_known_cost": 7542, "cost": 8785, "expected_gap": 0.237364, "family": "berlin", "name": "berlin52", "optimality_gap": 0.16481, "residual_gap": -0.072554}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.141226, synthetic 0.0, combined 0.089871.
- Accepted-epoch count 2, mean accepted code novelty 0.853725, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.186949, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.236357, mean selected residual gap 0.014864.
- Adaptation efficiency 0.044302 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.141226 across 7 instances; family means: ch=0.181646, kroD=0.139852, pcb=0.174052, pr=0.044388, rd=0.124779, st=0.142222.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3629, "expected_gap": 0.401939, "family": "a", "name": "a280", "optimality_gap": 0.407135, "residual_gap": 0.005196}, {"best_known_cost": 7542, "cost": 9857, "expected_gap": 0.237364, "family": "berlin", "name": "berlin52", "optimality_gap": 0.306948, "residual_gap": 0.069584}, {"best_known_cost": 629, "cost": 752, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.195548, "residual_gap": -0.060732}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.084761, synthetic 0.003941, combined 0.055372.
- Accepted-epoch count 3, mean accepted code novelty 0.817261, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.16044, size bias 8.3e-05, and failure concentration 0.157119.
- Replay selection diversity 0.271932, mean selected expected gap 0.156152, mean selected residual gap -0.0152.

- Adaptation efficiency 0.04933 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.084761 across 7 instances; family means: ch=0.066491, kroD=0.091528, pcb=0.203415, pr=0.034523, rd=0.037547, st=0.093333.
- Panel synthetic_holdout mean gap 0.003941 across 4 instances; family means: clustered_gaussian=0.015764, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3289, "expected_gap": 0.400853, "family": "a", "name": "a280", "optimality_gap": 0.275301, "residual_gap": -0.125552}, {"best_known_cost": 629, "cost": 686, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.09062, "residual_gap": -0.163434}, {"best_known_cost": 7542, "cost": 8088, "expected_gap": 0.233068, "family": "berlin", "name": "berlin52", "optimality_gap": 0.072395, "residual_gap": -0.160673}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.197162, synthetic 0.0, combined 0.125467.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.158073, mean hardness 0.331945, size bias 0.148439, and failure concentration 0.367262.
- Replay selection diversity 0.14525, mean selected expected gap 0.290998, mean selected residual gap 0.034844.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.197162 across 7 instances; family means: ch=0.106733, kroD=0.360618, pcb=0.231104, pr=0.256363, rd=0.213401, st=0.105185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3620, "expected_gap": 0.399923, "family": "a", "name": "a280", "optimality_gap": 0.403645, "residual_gap": 0.003722}, {"best_known_cost": 14379, "cost": 17908, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.245427, "residual_gap": -0.076084}, {"best_known_cost": 629, "cost": 678, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.077901, "residual_gap": -0.184102}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.086663, synthetic 0.0, combined 0.055149.
- Accepted-epoch count 3, mean accepted code novelty 0.814708, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.230022, size bias 0.120322, and failure concentration 0.174426.
- Replay selection diversity 0.196176, mean selected expected gap 0.308952, mean selected residual gap -0.071071.
- Adaptation efficiency 0.059496 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 1, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.086663 across 7 instances; family means: ch=0.094119, kroD=0.067061, pcb=0.177498, pr=0.058063, rd=0.083186, st=0.032593.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3435, "expected_gap": 0.398294, "family": "a", "name": "a280", "optimality_gap": 0.331912, "residual_gap": -0.066382}, {"best_known_cost": 14379, "cost": 18335, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.275123, "residual_gap": -0.046388}, {"best_known_cost": 629, "cost": 764, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.214626, "residual_gap": -0.052465}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.134684, synthetic 0.003941, combined 0.087141.
- Accepted-epoch count 2, mean accepted code novelty 0.645429, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.3043, size bias 0.120322, and failure concentration 0.17767.
- Replay selection diversity 0.205344, mean selected expected gap 0.31085, mean selected residual gap 0.004015.
- Adaptation efficiency 0.049276 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.134684 across 7 instances; family means: ch=0.160789, kroD=0.131492, pcb=0.224822, pr=0.091698, rd=0.068015, st=0.105185.
- Panel synthetic_holdout mean gap 0.003941 across 4 instances; family means: clustered_gaussian=0.015764, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3604, "expected_gap": 0.39969, "family": "a", "name": "a280", "optimality_gap": 0.397441, "residual_gap": -0.002249}, {"best_known_cost": 14379, "cost": 19202, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.33542, "residual_gap": 0.014146}, {"best_known_cost": 7542, "cost": 9685, "expected_gap": 0.238849, "family": "berlin", "name": "berlin52", "optimality_gap": 0.284142, "residual_gap": 0.045293}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Results

Key Metrics (Main Evidence for Transfer)

- **Held-out TSPLIB Gap** (mean_optimality_gap on heldout TSPLIB instances)
- **Synthetic Holdout Gap**
- **Transfer Gap**

Condition	Held-out TSPLIB Gap	Synthetic Holdout Gap	Transfer Gap	Archive Diversity	Archive Hardness	Archive Size Bias	Failure Concentration	Replay Mode	Selection Mode	Code Novelty (mean)	Notes
phase6_no_replay	0.086291	0.010343	0.058674	0.240179	0.188586	0.0	0.0	none	score_only	0.764505	Baseline; moderate gaps; no replay selection diversity or failure concentration

Condition	Held-out TSPLIB Gap	Synthetic Holdout Gap	Transfer Gap	Archive Diversity	Archive Hardness	Archive Size Bias	Failure Concentration	Replay Mode	Selection Mode	Code Novelty (mean)	Notes
phase6_random_replay	0.210832	0.012044	0.138545	0.236772	0.153461	8.3e-05	0.19457	random	score_only	0.0	Poor transfer & tsplib gaps; no code novelty; moderate failure concentration
phase6_failure_replay	0.143482	0.0	0.091307	0.174718	0.319117	0.124979	0.23356	failure	score_only	0.0	Best synthetic gap; improved transfer vs random replay; higher hardness and size bias
phase6_random_replay_compression	0.114249	0.0	0.072704	0.236772	0.188799	0.0	0.19457	random	novelty_gate	0.805692	Compression active; intermediate transfer; high code novelty with similar archive diversity
phase6_stratified_random_replay	0.141226	0.0	0.089871	0.240179	0.209654	0.0	0.221162	stratified_random	score_only	0.0	Slightly worse than failure replay on transfer; moderate failure concentration
phase6_diversity_weighted_replay	0.084761	0.003941	0.055372	0.240179	0.172416	0.0	0.105	diversity_weighted	score_only	0.817261	Best held-out TSPLIB gap (0.08476); best synthetic gap near zero; best transfer gap also near top
phase6_residual_failure_replay	0.197162	0.0	0.125467	0.18975	0.336098	0.127321	0.367262	residual_failure	score_only	0.0	Worst transfer & tsplib gaps; highest hardness and failure concentration
phase6_diversity_failure_replay	0.086663	0.0	0.055149	0.174718	0.263054	0.124979	0.174426	diversity_failure	score_only	0.814708	Best transfer gap (0.05515); near best held-out TSPLIB gap; reasonable hardness and failure concentration
phase6_diversity_failure_replay_compression	0.134684	0.003941	0.087141	0.174718	0.348079	0.124979	0.17767	diversity_failure	novelty_gate	0.645429	Compression on diversity_failure; transfer gap worse than non-compressed variant

Interpretation:

Transfer Performance

- ****Top Transfer Performance:**** Achieved by phase6_diversity_failure_replay (mean transfer gap 0.05515) and phase6_diversity_weighted_replay (0.05537), both significantly better than random or no-replay baselines.
- ****Synthetic Holdout Gap:**** Best synthetic gaps are near zero for phase6_failure_replay, phase6_diversity_weighted_replay, and compressed variants of diversity replay, implying better generalization to synthetic holdout distribution compared to random replay.
- ****Held-out TSPLIB Gap:**** phase6_diversity_weighted_replay achieves best held-out TSPLIB gap (0.08476), slightly better than no replay and diversity_failure replay.

Replay Archive Characteristics (Mechanism Studies)

- **Archive Diversity:** Highest in no_replay, random replay with compression, and diversity-weighted replay (~0.24); lower in failure replay and diversity_failure variants (~0.17).
- **Archive Hardness:** Failure replay and residual_failure replay have higher hardness (~0.32–0.34). Diversity_failure replay adds hardness (~0.26).
- **Size Bias:** Failure replay and related diversity_failure variants have non-zero positive archive size bias (~0.12); others near zero.
- **Failure Concentration:** Residual_failure replay has highest (0.367), failure replay 0.234, diversity_failure replay ~0.17–0.19, random replay ~0.19, and no replay 0.0.

Code Novelty vs Transfer

- **Code Novelty:** Lowest (0.0 mean) in replay modes involving failure or random replay without novelty gate.
- **High Code Novelty:** Observed with novelty gate and diversity-weighted replay (around 0.76–0.85).
- **Effect:** Despite code novelty drop in failure replay, transfer improves. Hence, better transfer is not due to increased code novelty but replay strategy.

Replay Modes Comparison

- **No Replay ("none"):** Baseline has moderate transfer and held-out gaps.
- **Random Replay:** High transfer and optimality gaps, poor transfer.
- **Failure Replay:** Improves synthetic holdout gap to zero, better transfer but higher hardness and archive size bias.
- **Diversity Weighted Replay:** Achieves best held-out and synthetic holdout gaps with moderate hardness and low failure concentration.
- **Diversity Failure Replay:** Best transfer, balanced hardness, and failure concentration.
- **Residual Failure Replay:** Worst transfer and held-out gaps with highest hardness and failure concentration.
- **Compression-aware Replay:** Slight decrease in transfer over non-compressed variants but increase in code novelty.

Overall Conclusions

- **Phase6 Diversity Weighted Replay** and **Phase6 Diversity Failure Replay** conditions achieve the best balance between transfer performance and solver optimality gaps, supported by low held-out TSPLIB and synthetic holdout gaps.
- Improved transfer does **not** correlate systematically with code novelty; failure-related replay modes lowered code novelty but improved transfer.
- Failure replay modes enhance archive hardness and concentrate on difficult instances but tend to increase failure concentration and size bias.
- Compression-aware variants reduce synthetic gap but at some cost to transfer performance.
- Random replay lacks transfer effectiveness despite moderate archive diversity.

Recommendations

- Prioritize **diversity weighted replay** or **diversity failure replay** for benchmark scenarios aiming at optimal balance of transfer accuracy and archive quality.
- Avoid pure random replay due to poor transfer and large optimality gaps.
- Leverage failure replay archives to improve synthetic holdout performance but monitor for failure concentration bias effects.
- Use novelty gate selectively; increases code novelty but does not always improve transfer metric.
- Further analysis on archive diversity and failure concentration dynamics can refine replay sample selection.